GAMIFIED VR EDUCATIONAL PLATFORM FOR MIDDLE SCHOOL CURRICULUM IN SRILANKA - SCIENCE MODULE WITH HISTORY

25-26J-423

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in Interactive Media

Department of Information Technology

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DECLARATION OF THE CANDIDATE & SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Wijenarayana C.H	IT22132314	

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

....

Signature of supervisor — Date: 2025/08/31

Mr. Didula Chamara Thanaweera Arachchi

....

Signature of co-supervisor — Date: 2025/08/31

Mr. Nushkan Nismi

ABSTRACT

The proposed project aims to develop an immersive Science Virtual Reality (VR) module for middle school education in Sri Lanka, as part of a gamified multi-subject VR learning platform. The module focuses on Grades 7–9 science topics—ecosystems, energy and simple machines, properties of matter, and plant biology—which often lack practical, hands-on learning opportunities. Current instruction relies heavily on textbooks and rote memorization, resulting in abstract conceptual understanding and reduced student engagement. To address this, the Science VR module will provide interactive laboratories and simulations, enabling students to conduct experiments safely and explore scientific phenomena that are otherwise infeasible in conventional classrooms. Developed using Unity for deployment on the Meta Quest 3 headset, the module includes high-fidelity 3D environments, natural interaction with objects, bilingual narration (Sinhala/English), real-time feedback, and gamified quest structures. A key innovation is the integration of science learning within historically themed VR worlds, situating experiments in local cultural contexts such as ancient irrigation systems or colonial-era machinery. This design promotes cross-curricular learning and enhances relevance for Sri Lankan students. The proposal outlines the research gap, objectives, methodology, and evaluation plan, including iterative development and pilot testing in local schools. The anticipated outcome is a curriculum-aligned VR science module that improves conceptual understanding, motivation, and engagement among middle school learners. By addressing both pedagogical and contextual challenges, the project aims to deliver a novel, scalable model for VR-based science education in Sri Lanka.

Keywords: Virtual Reality; Science Education; Gamification; Bilingual Learning; Sri Lanka.

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LIST OF ABBREVIATIONS

VR	Virtual Reality
AR	Augmented Reality
XR	Extended Reality (umbrella term for VR, AR, MR)
MR	Mixed Reality
HMD	Head-Mounted Display
UI	User Interface
UX	User Experience
SDK	Software Development Kit
API	Application Programming Interface
GPU	Graphics Processing Unit
FPS	Frames Per Second
LKR	Sri Lankan Rupees
SLIIT	Sri Lanka Institute of Information Technology

1 INTRODUCTION

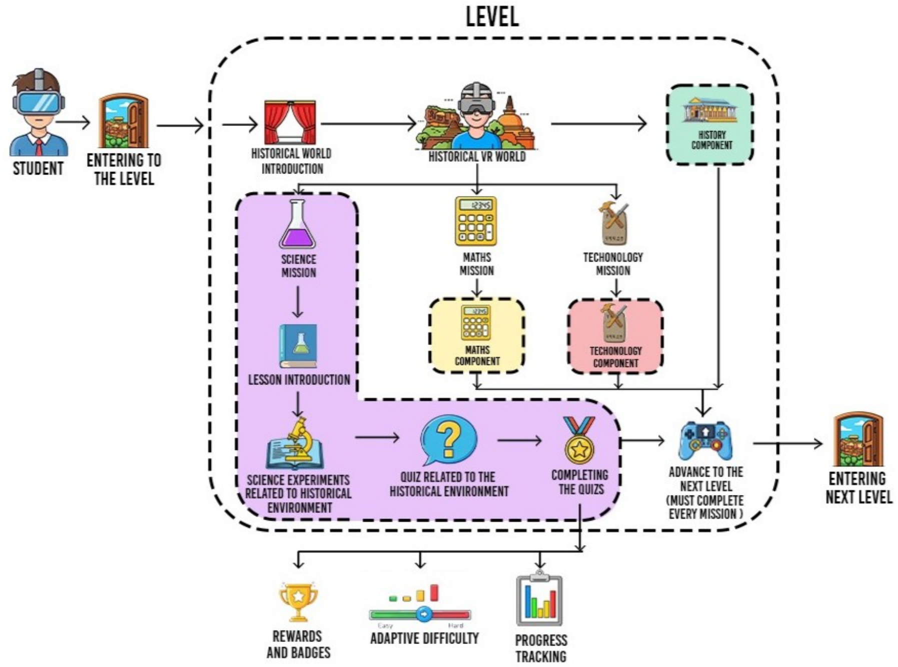

Figure 1.1: System overview of the proposed gamified VR science learning module.

1.1 Background & Literature Survey

Science education at the middle-school level in Sri Lanka faces significant challenges in delivering practical learning experiences. Although the national curriculum for Grades 7–9 covers fundamental topics in biology, physics, and chemistry, many schools – especially in rural or under-resourced areas – lack laboratory facilities and rely on didactic teaching. This reliance on textbook-based instruction and memorization leads to science concepts remaining abstract for students. Prior work on Sri Lankan education also notes similar systemic challenges in adopting effective e-learning practices [1]. For example, concepts such as ecosystem interactions, energy transformations, or chemical properties of matter are taught theoretically, with few opportunities for learners to observe or experiment firsthand. As a result, students often struggle to grasp these foundational science topics deeply and to retain interest in STEM subjects.

According to an informal survey of two local schools (Halwathura K.V. and Sri Dharmaloka K.V.), a majority of students reported that they rarely or never performed actual experiments for topics like plant physiology or simple machines. Many expressed difficulty understanding processes like photosynthesis or electrical circuits solely from diagrams and descriptions. This gap in experiential learning contributes to lower exam performance and reduced enthusiasm for science. It underscores a clear need for interactive and engaging learning tools that can simulate hands-on science learning in the absence of real labs.

Immersive technologies such as Virtual Reality (VR) offer a promising solution to this problem. VR can place students inside simulated scientific environments where they can learn by doing in a safe, controlled setting. Prior research has shown that VR-based learning can significantly improve engagement and conceptual understanding in science education [2, 3]. Students are able to visualize and manipulate 3D models of scientific phenomena – for instance, exploring a virtual ecosystem or assembling a virtual electrical circuit – leading to more concrete understanding of otherwise abstract concepts. Gamification elements (such as challenges, points, and narrative goals) further enhance engagement by framing learning as an interactive experience rather than a passive lesson [4].

Crucially, any educational intervention in Sri Lanka must align with the local curriculum and language context. Sri Lankan students are taught science in both Sinhala and English mediums, yet many struggle with English terminology. An effective solution should therefore offer bilingual support so that students can learn complex science concepts in their native language (Sinhala) while also building English vocabulary. Moreover, existing digital learning resources in Sri Lanka are scarce and often not well-integrated into formal curricula [1]. VR educational content, where available internationally, is rarely tailored to the Sri Lankan syllabus or available in Sinhala. This project addresses that gap by developing a VR science learning module explicitly mapped to the national Grades 7–9 science curriculum and supporting dual-language instruction.

The Science VR module is part of a larger gamified VR platform that integrates multiple subjects (History, Math, Science, and English) into one immersive experience. The group project's overarching concept uses Sri Lankan historical settings as a unifying narrative: students travel through virtual recreations of historical eras and encounter subject-specific learning quests embedded in each era's storyline. Within this context, the Science module contributes by providing interactive science experiments and simulations that complement the historical narrative.

Related Work and Novelty: Several existing VR applications demonstrate the potential of virtual science learning, but also reveal the gap that our project will fill. Labster, a well-known commercial platform, offers virtual science labs for high schools and universities worldwide [5]. However, Labster's content is oriented towards a global audience and higher grade levels, aligned with curricula like IB or AP, and delivered only in English. FotonVR provides immersive experiences aligned with international curricula, especially in biology and chemistry [6], while VRLab Academy enables learners to simulate experiments remotely on any device [7]. Yet, none of these existing solutions are customized for Sri Lankan students or incorporate local historical context or cross-curricular integration – a distinctive feature of our project.

1.2 Research Gap

Despite growing evidence of VR's effectiveness in education [2, 3], there is a clear gap when it comes to localized, curriculum-aligned VR content for Sri Lankan middle schools. Most VR-based science learning tools and studies to date focus on either higher education or generic science curricula in developed countries [8]. In Sri Lanka, to the best of our knowledge, no VR science modules exist that are specifically tailored to the national Grade 7–9 science syllabus or that cater to the bilingual (Sinhala–English) educational context. Furthermore, current science teaching in local schools rarely utilizes VR or any advanced interactive technology – conventional methods prevail, especially outside of elite urban schools [1].

Another aspect of the gap is the integration of subjects: while cross-curricular learning is encouraged by educators, existing VR applications tend to focus on a single subject [5–7]. Our interdisciplinary approach (embedding science in historical scenarios and gamifying it alongside math and language tasks) is novel, and there is little precedent or available product that offers this in the local context.

In summary, the gap can be described as: no existing solution provides an interactive VR science learning experience designed specifically for Sri Lankan middle school curricula (Grades 7–9), delivered in local languages with appropriate cultural context, and integrated into a multi-subject educational game. This project aims to fill that gap by developing and evaluating a prototype that meets all those criteria.

Table 1.1: Comparison of Existing Research and Proposed Module Contribution

Research Name	**Current Development (What they researched / where they stopped)**	**Future & Special Features (Our Module Contribution)**
Virtual Labs in Science Teaching: Sri Lankan Teachers' Perspectives and Challenges	Found VR labs boost engagement; focused on teacher views, not classroom roll-out.	Deliver offline VR labs with Sinhala–English support, gamification, cross-subject links, and full syllabus alignment.
Transforming Science Education with VR: Immersive Representations Model (IRM)	Proposed an evaluation framework; no practical classroom implementation.	Build hands-on VR lessons with Sinhala–English UI/voice, gamified paths, cross-curricular tasks, and offline use.
Virtual Reality in Science Education: A Descriptive Review	Positive results mainly in higher ed; weak school-level curriculum integration.	Provide school-level, syllabus-mapped VR science with Sinhala–English support, gamification, and offline access.
Adding Immersive VR to a Science Lab Simulation (Presence↑, Learning↓)	VR raised presence but reduced learning vs desktop; lacks pedagogy fixes.	Improve outcomes via guided interactions, Sinhala–English explanations, analytics, and mission-based design.
Bridging Gaps in Science Education with Virtual Labs	Noted internet/teacher-training barriers; theory only.	Ship offline VR labs + Sinhala–English teacher guides, gamified student tasks, and curriculum alignment.

Table 1.2: Feature Comparison of Existing VR Solutions vs Proposed Solution

Feature	Veative (STEM VR)	VictoryXR Academy	Prisms VR (STEM)	Kai XR	Titans of Space PLUS	Proposed Solution
Syllabus alignment to Sri Lankan Grades 7–9	✗	✗	✗	✗	✗	✓
Integration of Math, Science, English tasks inside history scenes	✗	✗	✗	✗	✗	✓
Bilingual narration (Sinhala & English)	✗	✗	✗	✗	✗	✓
Scene length optimized for 5–10 min missions	✓	✓	✓	✓	✓	✓
Gamified progression with rewards	✓	✓	✓	✗	✗	✓

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Table 1.2 (continued)

Feature	Veative (STEM VR)	VictoryXR Academy	Prisms VR (STEM)	Kai XR	Titans of Space PLUS	Proposed Solution
VR optimized for Meta Quest 3	✓	✓	✓	✓	✓	✓
Authentic Sri Lankan historical events	✗	✗	✗	✗	✗	✓

1.3 Research Problem

Sri Lankan middle-school students often struggle to engage with and fully understand key science concepts – such as ecological relationships, principles of energy and simple machines, properties of matter, and plant biology – due to the lack of practical experimental learning in the current education system.

Problem Statement: How can we improve conceptual understanding and student engagement in these Grade 7–9 science topics by designing a gamified VR-based learning module that is aligned with the national science curriculum and integrated into a historical narrative, while also being accessible in both Sinhala and English for learners in Sri Lanka?

This research problem addresses both the pedagogical challenge (enhancing understanding and engagement through interactivity and immersion) and the contextual challenge (alignment with local curriculum and language). The problem also implies several sub-questions:

- What design features are necessary in a VR science module to ensure it produces learning gains equivalent to or better than real labs?
- How can historical context and storytelling be used to enhance science learning without causing cognitive distraction?

- What are the measurable effects of using such a VR module on student motivation and academic performance in science?

2 OBJECTIVES

2.1 Main Objective

Develop an interactive, gamified Science VR module for the "Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka" that enables Grade 7–9 students to perform virtual science experiments and quests aligned to their curriculum, thereby improving understanding and engagement. The module shall integrate seamlessly into the platform's historical VR worlds and support bilingual (Sinhala/English) learning to serve under-resourced school environments.

2.2 Specific Objectives (SMART)

1. **Needs Analysis & Content Selection (Month 1).** Conduct a detailed review of the Grade 7–9 science syllabi and survey/interview at least 5 science teachers and 50 students (e.g., Halwathura K.V., Sri Dharmaloka K.V.) to identify *at least four* topic areas and practicals that are difficult or lack lab access. *Measurable:* four topics identified.
2. **Curriculum Mapping & Design Specification (Month 2).** Map each selected topic (e.g., G8 ecosystems/food webs; G9 energy/simple machines; G7 matter/states & changes; G7/8 plant biology/processes) to national learning outcomes. Produce a design document and storyboards specifying how each VR quest teaches those outcomes. *Measurable:* documented alignment per topic.
3. **VR Science Module Development (Months 3–6).** Implement a suite of *four* interactive simulations (mini-labs/quests) in Unity/C#, including immersive 3D scenes, guided Sinhala/English narration, interactive steps, and gamified elements (points, badges, and timers). Use physics/animation for realism (e.g., simple machines, water cycle, photosynthesis). Integrate technically with the platform launcher. *Measurable:* four functional simulations complete by Month 6.
4. **Integration with Historical Narrative (Month 7).** Collaborate with the History module owner (Jayalath K.M.K.D) to embed the science quests into *two* historical worlds (e.g., Anuradhapura, Colonial Ceylon) with smooth scene

transitions. *Measurable:* two integrations tested end-to-end.

5. **Pilot Deployment & User Training (Month 8).** Deploy the prototype to *two* schools; set up Meta Quest 3 devices; train ~2 teachers and ~30 students; run supervised sessions using the science quests. *Measurable:* 2 schools, ~30 students.
6. **Evaluation of Learning Outcomes & Engagement (Months 8–9).** Run pre/post tests per topic; target ≥20% average improvement in post-test scores. Collect analytics (time in VR, completion rates, quest scores) and gather teacher/student feedback on motivation and ease of learning. *Measurable:* score lift, analytics, survey metrics.
7. **Iteration & Finalization (Months 10–11).** Refine difficulty, narration clarity, performance on Quest, and usability issues found in the pilot; resolve known bugs. Compile the final report and documentation. *Measurable:* at least one iteration cycle completed; final deliverables submitted.

2.3 Milestones and KPIs

Table 2.1: Objectives, milestones, and key performance indicators (KPIs).

ID	Milestone / Deliverable	KPI / Success Metric	Target
O1	Needs analysis report identifying four priority topics	4 topics selected; 5 teachers and 50 students surveyed	M1
O2	Curriculum mapping and design storyboards completed	Each quest mapped to outcomes; design document approved	M2
O3	Development of four VR science simulations	Four quests functional with core features implemented	M6
O4	Integration of simulations into two historical VR worlds	Seamless launch and return in both scenarios	M7
O5	Pilot deployment in two schools with teacher/student training	Two schools reached; ~30 students engaged	M8
O6	Evaluation of learning outcomes and engagement	≥ 20% improvement in post-test scores; analytics collected	M9
O7	Iterated release and final project report	Issues resolved; final documentation submitted	M10–M11

3 METHODOLOGY

To accomplish the objectives, the project follows a **design-based research methodology** with iterative development and evaluation. This approach is appropriate for educational technology innovations, as it allows continuous refinement of the intervention (VR module) based on feedback from real users and alignment with theoretical principles. The development process is structured into four phases:

3.1 Phase 1: Requirements Gathering & Conceptual Design (Months 1–2)

- Review Grade 7–9 national science curriculum documents to extract specific learning outcomes and recommended practicals (e.g., Grade 7 plant parts; Grade 8 ecosystems; Grade 9 simple machines).
- Conduct surveys and interviews with teachers and students (Halwathura K.V., Sri Dharmaloka K.V.) to identify difficult or inaccessible topics.
- Select four key topics (see Objective 1) and perform a literature scan of VR/multimedia approaches to inspire design.
- Prepare detailed **storyboards** for each topic, showing the learner's journey in context (e.g., virtual paddy field in the Polonnaruwa era → investigate growth factors → earn points for optimizing photosynthesis).
- Outline **gamification features** (points, badges, narrative rewards) to motivate learners.
- Produce a conceptual system diagram linking the VR application, science module, gamification engine, and analytics system.

Output: Design specification, storyboards, list of required 3D assets, integration requirements.

3.2 Phase 2: Development & Integration (Months 3–7)

- **3D Environment & Assets:** Optimized low-poly models; blend custom Sri Lankan context assets (e.g., irrigation canals) with repository items.
- **VR Interactions:** Unity XR Toolkit + Oculus SDK for grabbing/placing, levers,

buttons; physics-based behavior (e.g., lever moments/forces).

- **Narration/UI:** Sinhala/English voice-overs; minimal VR-friendly UI with subtitles/tooltips; language toggle.
- **Gamification:** Points, badges, timers; local progress tracking (JSON/lightweight DB) for feedback and analytics.
- **History Integration:** Define triggers/scene transitions (e.g., Anuradhapura tank $\rightarrow$ ecosystem simulation) with consistent art style.
- **Internal Testing:** Meta Quest 3 tests for legibility, intuitive controls, comfort; quick fixes each mini-cycle.

Output: Integrated Science module with four simulations accessible via the platform.

3.3 Phase 3: Pilot Testing & Data Collection (Month 8)

- Deploy prototype to Halwathura K.V. and Sri Dharmaloka K.V.; obtain permissions/consent.
- Train science teachers on device handling and session flow.
- Run student sessions in rotations (1–2 Quest devices; cast to monitor for observers); record observations (immersion, difficulties, reactions).
- Administer pre/post tests per topic; collect student surveys (Likert + open-ended) and teacher interviews.
- Log analytics in-app (time on task, completion rate, quest scores).

Output: Dataset of scores, surveys, logs, and teacher feedback.

3.4 Phase 4: Evaluation & Iteration (Months 9–10)

- **Learning Outcomes:** Analyze pre/post improvements (paired t-test where applicable); target $\geq 20\%$ average gain.
- **User Experience:** Summarize survey ratings and themes; identify pain points (confusion, motion discomfort).
- **Teacher Feedback:** Assess practicality (time fit, curriculum tie-ins); capture integration suggestions.
- **Refinement:** Tune difficulty, narration clarity, guidance cues; optimize perfor-

mance and comfort; fix bugs; prepare final docs.

Output: Polished Science VR module, final report, and teacher guide.

3.5 Ethical Considerations & Dissemination

Permissions will be obtained from schools, parents, and participants, ensuring student safety and informed consent. The final results will be shared with stakeholders and integrated with the group's other subject modules (History, Math, English). The methodology provides a pilot model for scaling VR-based education across more schools in Sri Lanka.

3.6 Functional Requirements

FR1: **Virtual Experiment Simulations** – Provide interactive simulations for at least four Grade 7–9 science topics (ecosystems, simple machines, matter, plant biology). Each simulation should replicate real-world experiments (e.g., mixing virtual chemicals, assembling machine parts).

FR2: **Guided Instruction and Narrative** – Present each activity with Sinhala/English audio narration and contextual storytelling. Narration must explain steps, pose inquiry prompts, and connect to the historical storyline.

FR3: **User Interaction with Objects** – Support natural VR interactions (grabbing, pouring, combining, activating objects). Physics must be realistic (objects fall, levers respond to force, etc.).

FR4: **Immediate Feedback and Experiment Outcomes** – Provide visual/auditory feedback for correct and incorrect actions. Reinforce learning with explanations and a short summary screen/narration after each activity.

FR5: **Gamification and Scoring** – Award points, badges, and progress levels. Badges (e.g., "Eco Warrior," "Lab Technician") tie into the wider platform progression system.

FR6: **Progress Tracking** – Record metrics such as completion, time taken, hints used, and scores. Export data for teachers/researchers; integrate with platform analytics.

FR7: **Bilingual Support** – Deliver all instructions, narration, and UI in both

Sinhala and English. Support dual-language display where needed.

FR8: **Module Integration** – Launch and exit seamlessly from the multi-subject platform. Support state handover (e.g., trigger from a history quest $\rightarrow$ return to main storyline).

3.7 Non-Functional Requirements

NFR1: **Usability and Child-Friendly Design** – Simple, intuitive controls suitable for ages 12–15. Provide visual hints, large legible fonts, and options to repeat narration or quit at any time.

NFR2: **Performance and Comfort** – Maintain $\geq$72 FPS on Meta Quest 3; load times under 5s. Follow VR comfort best practices (stable horizon, teleportation instead of large walking).

NFR3: **Safety** – Ensure safe VR use (guardian boundaries, headset strap warnings). Simulated labs must model safe practices and prevent dangerous outcomes.

NFR4: **Localization and Cultural Relevance** – Use Sri Lankan context (local fauna/flora, history, attire, narration tone). Content must be accurate, respectful, and age-appropriate.

NFR5: **Maintainability and Modularity** – Structure Unity scenes/prefabs modularly, with documentation and developer notes for easy extension or updates.

NFR6: **Scalability** – Support scaling from one device to multiple headsets. Prepare architecture for future multi-user or expanded Grade 10+ content.

NFR7: **Compatibility** – Primary target is Meta Quest 3; ensure portability to Quest 2 and other XR devices. Optionally support a 2D demo/spectator mode for teachers.

NFR8: **Evaluation and Logging** – Log usage data (timestamps, actions, errors, performance metrics) for debugging and research evaluation, with proper consent.

4 DESCRIPTION OF PERSONNEL AND FACILITIES

4.1 Project Personnel

This project will be carried out by **C.H. Wijenarayana** (Student, IT22613942) as part of the requirements for the *B.Sc. (Hons) in Information Technology* degree, under the supervision of **[Supervisor's Name]** from the Faculty of Computing, SLIIT.

The student specializes in Interactive Media and has a background in Unity 3D development, game design, and user experience design through coursework and prior projects. The supervisor will provide guidance on research methodology, technical validation, and educational design principles throughout the project.

Additional support will come from team members in the larger group project—each focusing on different subject modules (History, Math, and English)—ensuring cross-module integration. Input from subject matter experts (e.g., science teachers and curriculum specialists) will also be sought during requirements gathering and evaluation phases, though they are not formal project personnel.

The project will also involve collaboration with two pilot schools, **Halwathura K.V.** and **Sri Dharmaloka K.V.**, for needs analysis and testing. Teachers and administrators at these schools will act as stakeholders and facilitators, providing valuable feedback and logistical support.

4.2 Facilities and Resources

Development and testing will primarily take place at the **Faculty of Computing, SLIIT**. A high-performance development workstation (PC) is available in the university lab for running Unity and VR development tools. This workstation includes a dedicated GPU capable of handling 3D rendering and VR simulation during development. No additional PC purchase is required, as existing equipment will be utilized.

A **Meta Quest 3** VR headset (standalone VR device) will be used for building and testing the immersive experience under real conditions. This device will also serve as the primary hardware for demonstrations and pilot deployments, as its advanced tracking and standalone operation make it suitable for classroom use without requiring an external PC.

The software stack will include the Unity game engine (Personal/Education license) and freely available plugins such as the Unity XR Interaction Toolkit and Oculus Integration SDK. SLIIT provides access to this software environment for students.

The team also has access to additional facilities such as **usability labs** and study areas at SLIIT, which can be booked for internal testing sessions or to simulate classroom conditions for pilot preparation. For audio content creation, the university's multimedia studio or a quiet lab space with recording equipment will be used to record Sinhala and English narrations. If higher-quality recordings are required, an external studio may be rented (as budgeted).

During pilot testing at partner schools, basic facilities such as classrooms or computer labs will be used. The project will supply the VR headset and, if needed, a laptop for casting the VR view. Schools will provide space and supervision for student participation. Internet connectivity is not required for the VR module (content runs locally), but the university's network will be used for downloading assets, updates, or remote collaboration.

Summary

In summary, the personnel involved combine technical expertise (the student developer and supervisor), interdisciplinary support (peer team members), and domain guidance (teachers and curriculum experts). The facilities available—including computing labs, a VR headset, and pilot school venues—are sufficient to design, implement, and evaluate the Science VR module in realistic classroom-like environments. These resources collectively ensure a high likelihood of project success.

5 BUDGET AND BUDGET JUSTIFICATION

The implementation of the Science VR Module requires certain resources, both in terms of hardware and software, as well as operational expenses for research activities. Table 5.1 summarizes the estimated budget, followed by justification for each item.

Table 5.1: Estimated Budget for Science VR Module Development

Item / Resource	Quantity	Estimated Cost (LKR)	Justification
Meta Quest 3 VR Headset (256 GB)	1 unit	180,000	Primary hardware for development and demonstration. Essential for building and testing the immersive experience under real conditions. Includes controllers and standard warranty.
Development Workstation (PC upgrade)	– (in-kind)	0	Existing high-performance PC (lab or personal) will be used. If required, minor upgrades (e.g., RAM) will be covered by existing budgets.
Unity 3D Software & Plugins	–	0	Unity is free under Personal/Education license. Free plugins (Unity XR Interaction Toolkit, Oculus Integration SDK) will be used.

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Item / Resource	Quantity	Estimated Cost (LKR)	Justification
3D Asset Purchases (Unity Asset Store)	Lump sum	30,000	Budget for specialized 3D models or textures (e.g., anatomy model, detailed lab equipment). Estimated five packs (~USD 30–50 each). Unused funds may be reallocated.
Audio Recording (Narration)	Lump sum	10,000	Sinhala/English narration. May require renting a studio, equipment, or a stipend for voice actors. Budget ensures clear, accent-appropriate recordings.
Testing & Travel Expenses	–	15,000	Travel to pilot schools (Halwathura K.V., Sri Dharmaloka K.V.), including transport, refreshments, printing consent forms, and incidental costs for surveys, deployment, and follow-up.

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Item / Resource	**Quantity**	**Estimated Cost (LKR)**	**Justification**
Miscellaneous (Contingency 10%)	–	20,000	Covers unexpected costs (e.g., replacement VR controller ~15,000 LKR, spare batteries/cables, hygiene accessories).
Total Estimated Budget		**255,000**	*Approximately USD $750, assuming 1 USD ~ 340 LKR.*

Budget Justification. The largest cost driver is the Meta Quest 3 headset (180,000 LKR), which is indispensable for development, testing, and deployment in real classroom contexts. No cost is allocated for the workstation or software, since existing resources and free licenses are sufficient. A modest allocation of 30,000 LKR is provided for asset purchases to ensure high-quality and contextually accurate 3D models. Additional funds are budgeted for narration (10,000 LKR) to guarantee clear bilingual delivery, while 15,000 LKR covers transport and incidental costs of pilot testing in rural schools. Finally, a contingency of 20,000 LKR ensures resilience against unforeseen issues such as hardware replacement or minor usability improvements. Together, these allocations provide a lean yet complete budget that ensures technical feasibility, educational quality, and practical deployment of the Science VR module.

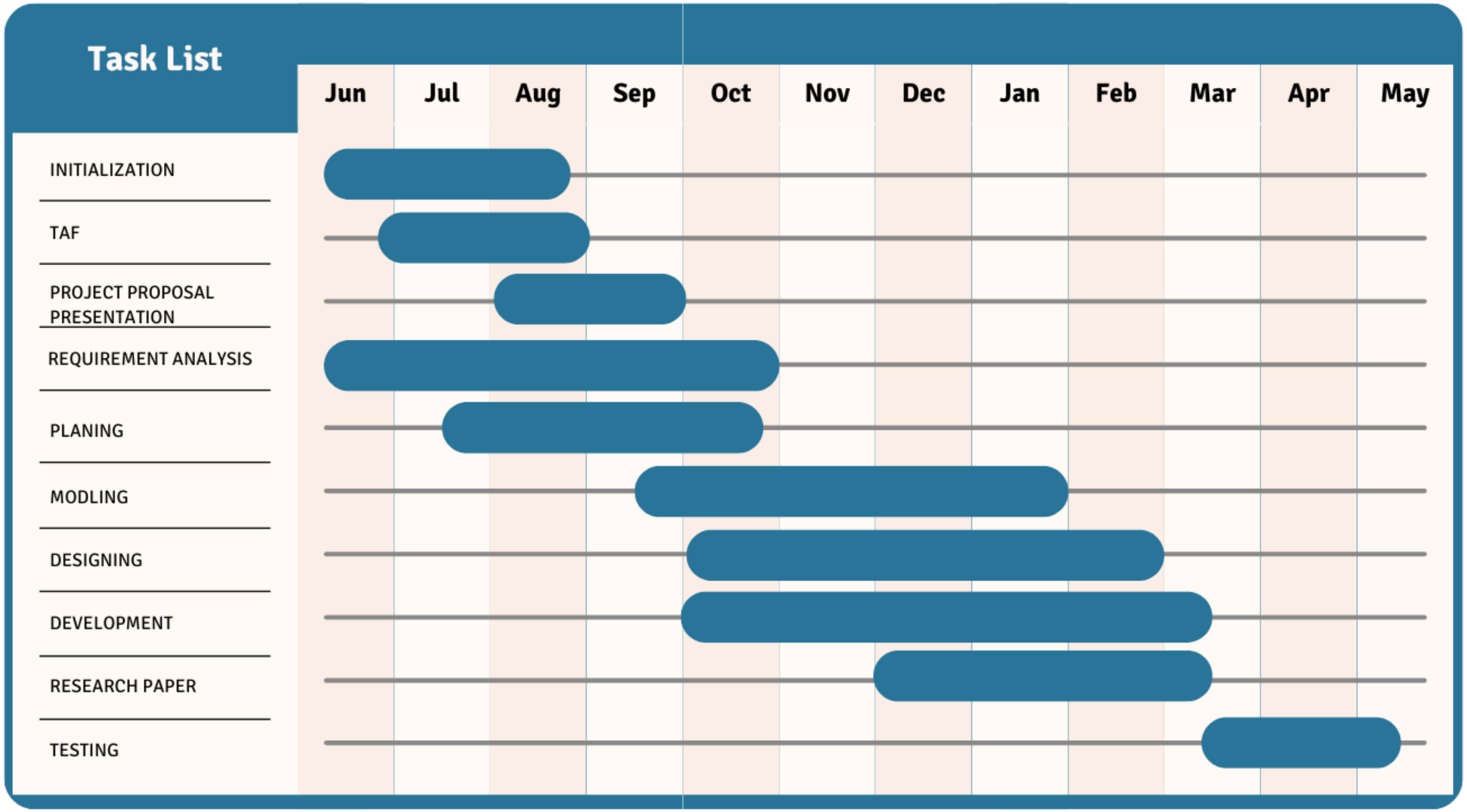

Figure 5.1: Project Gantt Chart for Science VR Module Development

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